

CUSUM Active-Manager Monitor

Statistical Process Control for Portfolio Manager Oversight

Report generated: 09 May 2026

What is CUSUM?

CUSUM (Cumulative Sum) is a sequential change-point detection method from statistical process control, adapted for investment management by Philips, Yashchin & Stein (2003). It answers one critical question:

Has a fund manager lost their investment skill — and if so, exactly when?

The Case for CUSUM — Why CIOs Should Care

Cost of Inaction:

Retaining an underperforming manager for 3 extra years on a 500M mandate at $IR = -0.3$, $TE = 6\%$ costs ~9M in excess losses. Early, systematic detection pays.

Faster than Rolling IR:

A trailing 3-year IR review needs 10+ years of data to distinguish skill from luck at 95% confidence. CUSUM detects persistent changes in IR within 2-3 years.

Rigorous False-Alarm Control:

Threshold h is calibrated from the paper's ARL table: $h = 19.81$ gives ~1 false alarm per 60 months (5 years) when the true $IR = +0.5$. Fewer spurious terminations.

Two-Sided Monitoring:

Detects skill loss (lower CUSUM) and unexpected skill gain (upper CUSUM), enabling both termination decisions and early-stage manager promotions.

Rolling Restart:

After each alarm L resets to zero — monitoring never stops. Multiple alarms build a richer signal; post-alarm returns validate each flag.

Actionable, Not Automatic:

An alarm triggers a structured review — not automatic termination. It forces the investment committee to ask: Is the alpha source still intact?

Executive Summary

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68

Funds
Monitored

51

CUSUM
Alarms

22

Require
Review

17

No Alarm
(Clean)

Performance Summary by Asset Class

Asset Class	Funds	Alarmed	Avg IR	Hit%	Best Fund	Signal
US Large Blend	3	3	-0.12	49%	AGTHX (+0.10)	REVIEW
US Large Growth	1	1	+0.04	47%	FCNTX (+0.04)	WATCH
US Large Value	1	1	+0.14	52%	DODGX (+0.14)	WATCH
US Small Cap	1	1	+0.32	54%	OTCFX (+0.32)	WATCH
US Small/Mid Cap	1	1	+0.09	50%	FLPSX (+0.09)	WATCH
Balanced 60/40	2	—	+0.80	63%	PRWCX (+0.97)	CLEAN
US Fixed Income	2	—	+0.36	62%	PTTRX (+0.50)	CLEAN
Intl Developed	1	1	+0.06	49%	OAKIX (+0.06)	WATCH
Intl Value	1	—	+0.42	56%	DODWX (+0.42)	CLEAN
Emerging Markets	27	23	+0.04	52%	FSEAX (+0.42)	REVIEW
EM Hard CCY Debt	11	7	-0.05	50%	IEMIX (+1.58)	REVIEW
EM Local CCY Debt	2	2	+0.15	49%	PFMIX (+0.26)	REVIEW
Global Equity	15	11	-0.33	45%	OP00008T6D.SI (+0.64)	REVIEW

Key Findings

- 51 of 68 funds (75%) triggered a CUSUM alarm between Jan 2005 and today.
- Regime alarms — GFC (Oct'07–Mar'09): 1 | COVID (Feb–Apr'20): 0 — these reflect market stress, not skill loss.
- 22 fund(s) classified REVIEW (confirmed by negative post-alarm return at +12M); 28 classified WATCH.
- Best risk-adjusted manager: IEMIX (IR +1.58, t=+0.9). Weakest: OP0001RMIN.SI (IR -1.46).
- Multiple testing: at ARL=60m per fund, expect ~13.6 false alarms/year across 68 funds — see Summary section for detail.

How CUSUM Works

Step-by-step mechanics of the Philips-Yashchin-Stein procedure

1

Step 1 — Log Excess Return

$$et = \ln[(1 + r_{fund}) / (1 + r_{benchmark})]$$

Log returns are additive and symmetric — they handle compounding correctly.

2

Step 2 — EWMA Tracking-Error Estimate

$$\sigma^2_t = \gamma * \sigma^2(t-1) + (1-\gamma) * 0.5*(et - e(t-1))^2 \quad [\gamma=0.90]$$

Von Neumann adjacent-difference estimator: robust to drift in the mean, adapts as the manager's risk budget changes.

3

Step 3 — Monthly IR Estimate

$$IR_{hat_t} = (et * \sqrt{12}) / \sigma(t-1) \quad [no\ clipping]$$

Annualised IR using lagged sigma. If |IR_hat| > regime_threshold (default 4.0), the month is a regime outlier: L and L_up are frozen (not accumulated) and the index is recorded. Cleaner than winsorisation — the raw IR is preserved.

4

Step 4 — 24-Month Calibration Gate

First 24 months: sigma EWMA runs; L frozen at 0.

Prevents in-sample contamination: the same data used to initialise sigma is not also tested against it.

5

Step 5 — Lindley Recursion (Page's CUSUM)

$$L_t = \max(0, L(t-1) + 0.25 - IR_{hat_t}) \quad [drift = 0.5*(IR_{good}+IR_{bad}) - IR_{hat}]$$

L grows when the manager underperforms, resets to 0 when outperforming. Upper CUSUM L_up uses drift_up = IR_hat - 0.5*(IR_exceptional + IR_good).

6

Step 6 — Rolling Alarm Rule

ALARM if L_t >= h = 19.81 then reset L_t = 0 and continue

After each alarm L resets to zero — monitoring never stops. All alarm times are recorded; post-alarm returns validate each signal.

Alarm Threshold Reference Table (Table 2, Philips et al. 2003)

Threshold (h)	ARL — Good (IR=0.5)	ARL — Flat (IR=0)	ARL — Bad (IR=−0.5)
11.81	24.00	16.00	11.00
15.00	36.00	22.00	15.00
17.60	48.00	27.00	18.00
19.81	60.00	32.00	21.00
21.79	72.00	37.00	23.00
23.59	84.00	41.00	25.00

Model Assumptions & Parameters

Philips-Yashchin-Stein (2003) — implementation reference

Algorithm Parameters

Parameter	Value	Rationale
Threshold (h)	19.81	ARL $\approx$ 60 months under H ₀ at IR = +0.5
EWMA decay (γ)	0.90	Von Neumann estimator; $\sim$ 10-month half-life
μ_{good} (IR floor)	+0.50	Signifies 'acceptable' manager skill
μ_{bad} (IR floor)	0.00	Signifies 'underperforming' manager
$\mu_{\text{exceptional}}$	+1.00	Upper-CUSUM trigger for skill gain
Calibration window	24 months	Sigma warmup only; L frozen at zero
Regime threshold	± 4.0	If $ \text{IR}_{\text{hat}} > 4$: freeze L, record outlier month
Bootstrap blocks	$n^{(1/3)}$	Circular block (Politis-White 2004)
Min. monitoring	36 months	Below this, results flagged * (low conf.)

Alarm Classification Guide

- REVIEW:** Alarm confirmed — fund underperformed at +12M. Schedule IC review.
- WATCH:** Alarm raised; inconclusive post-alarm data or fund recovered.
- REGIME:** Alarm in GFC or COVID window. Likely market-driven, not skill loss.
- CLEAN:** No alarm. Monitor at standard quarterly cadence.

Data Source & Coverage

- Source:

Yahoo Finance (yfinance) — no API key required
- Prices:

Total-return adjusted close (dividends reinvested)
- Frequency:

Daily prices $\rightarrow$ month-end to month-end returns
- Universe start:

January 2005 (common start for all funds)
- Returns used:

Log excess: $e_t = \ln[(1+r_{\text{fund}}) / (1+r_{\text{bench}})]$
- Benchmarks:

SPY · IWF · IWD · IWM · EEM · AGG · EMB · EMLC · ACWI · URTH · AOR

What an Alarm Means

A CUSUM alarm ($L \geq h$) indicates that accumulated evidence of underperformance is statistically significant. It should trigger:

- A structured investment-committee review
- A mandate compliance check with the portfolio manager
- NOT automatic termination

L resets to 0 after each alarm (rolling restart). Monitoring never stops.

Known Limitations

- Multiple testing: 67 funds $\times$ ARL=60m $\rightarrow \sim$ 13 false alarms/year across universe.
- Yahoo Finance adjusted prices may differ from official NAV series.
- Singapore AIAIM funds use Morningstar proxy IDs (OP...) with limited history.
- Regime outlier months ($|\text{IR}_{\text{hat}}| > 4$, e.g. March 2020): L frozen rather than clipped— no distortion.
- Calibration sigma uses first 24m of actual fund data (mild look-ahead).
- Single uniform threshold $h=19.81$; optimal h varies by mandate TE.

Fund Universe

68 active managers grouped by asset class — page 1 of 5

#	Ticker	Fund Name	Benchmark	Asset Class	h
■ US LARGE BLEND					
1	AGTHX	AmFunds Growth Fund of America vs S&P 500	SPY	US Large Blend	19.81
2	FMAGX	Fidelity Magellan vs S&P 500	SPY	US Large Blend	19.81
3	SEQUX	Sequoia Fund vs S&P 500	SPY	US Large Blend	19.81
■ US LARGE GROWTH					
4	FCNTX	Fidelity Contrafund vs Russell 1000 Growth	IWF	US Large Growth	19.81
■ US LARGE VALUE					
5	DODGX	Dodge & Cox Stock vs Russell 1000 Value	IWD	US Large Value	19.81
■ US SMALL CAP					
6	OTCFX	T. Rowe Price Small-Cap Stock vs Russell 2000	IWM	US Small Cap	19.81
■ US SMALL/MID CAP					
7	FLPSX	Fidelity Low-Priced Stock vs Russell 2000	IWM	US Small/Mid Cap	19.81
■ BALANCED 60/40					
8	PRWCX	T. Rowe Price Capital Appreciation vs 60/40 Allocation	AOR	Balanced 60/40	19.81
9	VWELX	Vanguard Wellington vs 60/40 Allocation	AOR	Balanced 60/40	19.81
■ US FIXED INCOME					
10	LSBRX	Loomis Sayles Bond vs US Agg Bond	AGG	US Fixed Income	19.81
11	PTTRX	PIMCO Total Return vs US Agg Bond	AGG	US Fixed Income	19.81
■ INTL DEVELOPED					
12	OAKIX	Oakmark International vs MSCI EAFE	EFA	Intl Developed	19.81

Fund Universe

68 active managers grouped by asset class — page 2 of 5

#	Ticker	Fund Name	Benchmark	Asset Class	h
■ INTL VALUE					
13	DODWX	Dodge & Cox International vs MSCI EAFE	EFA	Intl Value	19.81
■ EMERGING MARKETS					
14	ABEMX	AB EM Growth vs MSCI EM	EEM	Emerging Markets	19.81
15	BGEFX	Baron EM vs MSCI EM	EEM	Emerging Markets	19.81
16	EEMAX	Eaton Vance EM vs MSCI EM	EEM	Emerging Markets	19.81
17	FSEAX	Fidelity EM vs MSCI EM	EEM	Emerging Markets	19.81
18	GERIX	Goldman Sachs EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
19	HAEMX	Harbor EM vs MSCI EM	EEM	Emerging Markets	19.81
20	HLEMX	Harding Loevner EM Institutional vs MSCI EM	EEM	Emerging Markets	19.81
21	HLMEX	Harding Loevner EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
22	JEMWX	JPMorgan EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
23	LZOEX	Lazard EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
24	MADCX	Madison EM vs MSCI EM	EEM	Emerging Markets	19.81
25	MEMCX	Matthews EM vs MSCI EM	EEM	Emerging Markets	19.81
26	MIEMX	MFS EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
27	NEWFX	American Funds New World vs MSCI EM	EEM	Emerging Markets	19.81
28	NWFFX	New World Fund vs MSCI EM	EEM	Emerging Markets	19.81
29	ODMAX	Invesco Developing Markets vs MSCI EM	EEM	Emerging Markets	19.81
30	PDEZX	PGIM EM vs MSCI EM	EEM	Emerging Markets	19.81

Fund Universe

68 active managers grouped by asset class — page 3 of 5

#	Ticker	Fund Name	Benchmark	Asset Class	h
31	PEIFX	PGIM Jennison EM Equity vs MSCI EM	EEM	Emerging Markets	19.81
32	PRMSX	T. Rowe Price EM Stock vs MSCI EM	EEM	Emerging Markets	19.81
33	REMI	Royce EM vs MSCI EM	EEM	Emerging Markets	19.81
34	SFENX	Seafarer EM Growth & Income vs MSCI EM	EEM	Emerging Markets	19.81
35	TEDMX	Templeton Developing Markets vs MSCI EM	EEM	Emerging Markets	19.81
36	TEMF	Templeton EM vs MSCI EM	EEM	Emerging Markets	19.81
37	VEIE	Vanguard EM Stock Index vs MSCI EM	EEM	Emerging Markets	19.81
38	VEMAX	Vanguard EM Stock Index Admiral vs MSCI EM	EEM	Emerging Markets	19.81
39	WAEM	Western Asset EM vs MSCI EM	EEM	Emerging Markets	19.81
40	WIEM	William Blair EM Growth vs MSCI EM	EEM	Emerging Markets	19.81
■ EM HARD CCY DEBT					
41	0P00008T6N.SI	AIAIM EM Bond Fund B vs JPM EM Bond	EMB	EM Hard CCY Debt	19.81
42	0P0001KL95.SI	AIAIM EM Bond Fund A vs JPM EM Bond	EMB	EM Hard CCY Debt	19.81
43	ABEM	AB EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
44	DBLE	DFA EM Fixed Income vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
45	EBND	BlackRock EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
46	IEM	Invesco EM Sovereign Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
47	MSED	MFS EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
48	PEB	PIMCO EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
49	PREM	T. Rowe Price EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81

Fund Universe

68 active managers grouped by asset class — page 4 of 5

#	Ticker	Fund Name	Benchmark	Asset Class	h
50	TGBAX	Templeton Global Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
51	VWOB	Vanguard EM Govt Bond ETF vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	19.81
■ EM LOCAL CCY DEBT					
52	PFMIX	PIMCO Foreign Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	19.81
53	PRLAX	T. Rowe Price EM Local CCY Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	19.81
■ GLOBAL EQUITY					
54	0P00008T6D.SI	AIAIM Global Equity Fund D vs MSCI ACWI	ACWI	Global Equity	19.81
55	0P00008T6G.SI	AIAIM Global Equity Fund C vs MSCI ACWI	ACWI	Global Equity	19.81
56	0P0001I6KI.SI	AIAIM Global Equity Fund A vs MSCI ACWI	ACWI	Global Equity	19.81
57	0P0001RMIN.SI	AIAIM Global Equity Fund E vs MSCI ACWI	ACWI	Global Equity	19.81
58	AIVSX	American Investors vs MSCI World	URTH	Global Equity	19.81
59	CWGIX	Capital World Growth & Income vs MSCI ACWI	ACWI	Global Equity	19.81
60	GPROX	Goldman Sachs Global Equity vs MSCI ACWI	ACWI	Global Equity	19.81
61	HAIX	Harbor International vs MSCI World	URTH	Global Equity	19.81
62	JGVAX	JPMorgan Global Focus vs MSCI World	URTH	Global Equity	19.81
63	MGGIX	MFS Global Equity vs MSCI ACWI	ACWI	Global Equity	19.81
64	MGIAX	Morgan Stanley Institutional Global vs MSCI World	URTH	Global Equity	19.81
65	OPGIX	Invesco Oppenheimer Global vs MSCI ACWI	ACWI	Global Equity	19.81
66	SGIGX	Schroders Global vs MSCI ACWI	ACWI	Global Equity	19.81
67	TEDIX	Templeton Global Equity vs MSCI ACWI	ACWI	Global Equity	19.81

Fund Universe

68 active managers grouped by asset class — page 5 of 5

#	Ticker	Fund Name	Benchmark	Asset Class	h
68	VHGEX	Vanguard Global Equity vs MSCI ACWI	ACWI	Global Equity	19.81

Summary Results

Risk-adjusted statistics — grouped by asset class, 24m calibration excluded — page 1 of 5

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	First Alarm	Signal
■ US LARGE BLEND								
AGTHX	SPY	+0.10 [-0.37,+0.60]	+0.45	53%	103%/101%	+9.9%	Jan 2011	REVIEW
FMAGX	SPY	-0.22 [-0.71,+0.33]	-1.01	49%	104%/113%	-24.0%	Dec 2011	WATCH
SEQUX	SPY	-0.22 [-0.65,+0.23]	-1.03	46%	85%/86%	-33.2%	Sep 2018	WATCH
■ US LARGE GROWTH								
FCNTX	IWF	+0.04 [-0.35,+0.41]	+0.17	47%	93%/86%	+3.4%	May 2011	WATCH
■ US LARGE VALUE								
DODGX	IWD	+0.14 [-0.29,+0.60]	+0.66	52%	105%/103%	+14.5%	Dec 2011	WATCH
■ US SMALL CAP								
OTCFX	IWM	+0.32 [-0.14,+0.82]	+1.49	54%	92%/84%	+32.7%	Nov 2013	WATCH
■ US SMALL/MID CAP								
FLPSX	IWM	+0.09 [-0.29,+0.46]	+0.41	50%	77%/67%	+17.2%	Oct 2010	WATCH
■ BALANCED 60/40								
PRWCX	AOR	+0.97 [+0.66,+1.45]	+4.07	66%	111%/74%	+119.1%	-	CLEAN
VWELX	AOR	+0.64 [+0.39,+1.10]	+2.66	60%	106%/85%	+58.0%	-	CLEAN
■ US FIXED INCOME								
LSBRX	AGG	+0.22 [-0.29,+0.86]	+1.02	59%	112%/78%	+39.1%	-	CLEAN
PTTRX	AGG	+0.50 [+0.04,+1.02]	+2.30	64%	107%/90%	+22.2%	-	CLEAN

Summary Results

Risk-adjusted statistics — grouped by asset class, 24m calibration excluded — page 2 of 5

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	First Alarm	Signal
■ INTL DEVELOPED								
OAKIX	EFA	+0.06 [-0.37,+0.51]	+0.27	49%	111%/110%	+11.0%	Jul 2014	WATCH
■ INTL VALUE								
DODWX	EFA	+0.42 [-0.01,+0.92]	+1.79	56%	109%/95%	+72.3%	-	CLEAN
■ EMERGING MARKETS								
ABEMX	EEM	+0.24 [-0.21,+0.69]	+1.05	55%	95%/88%	+30.8%	May 2024	REVIEW
BGEFX	EEM	+0.05 [-0.62,+0.76]	+0.17	48%	99%/93%	+6.9%	Oct 2025	WATCH
EEMAX	EEM	+0.04 [-0.45,+0.55]	+0.19	51%	98%/96%	+4.7%	Dec 2018	WATCH
FSEAX	EEM	+0.42 [-0.07,+0.93]	+1.93	59%	101%/86%	+120.3%	Oct 2016	WATCH
GERIX	EEM	+0.23 [-0.20,+0.65]	+0.99	55%	101%/97%	+21.5%	Apr 2019	REVIEW
HAEMX*	EEM	-0.55 [-1.06,+0.95]	-1.09	55%	61%/103%	-39.3%	-	CLEAN
HLEMX	EEM	-0.42 [-1.04,+0.34]	-1.10	49%	100%/111%	-15.7%	Oct 2025	WATCH
HLMEX	EEM	-0.06 [-0.43,+0.33]	-0.29	50%	93%/92%	-7.1%	Aug 2018	WATCH
JEMWX	EEM	+0.39 [-0.24,+1.12]	+1.39	62%	105%/97%	+27.3%	-	CLEAN
LZOEX	EEM	+0.15 [-0.29,+0.58]	+0.71	50%	94%/88%	+23.4%	Aug 2017	REVIEW
MADCX	EEM	+0.12 [-0.23,+0.47]	+0.57	51%	98%/95%	+14.7%	Dec 2008	REGIME
MEMCX	EEM	-0.32 [-0.69,+0.01]	-1.49	50%	90%/93%	-26.9%	Oct 2013	REVIEW
MIEMX	EEM	+0.02 [-0.45,+0.51]	+0.09	51%	84%/78%	+2.3%	Sep 2012	REVIEW

Summary Results

Risk-adjusted statistics — grouped by asset class, 24m calibration excluded — page 3 of 5

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	First Alarm	Signal
NEWFX	EEM	+0.16 [-0.19,+0.52]	+0.73	51%	79%/68%	+29.4%	Mar 2025	REVIEW
NWFFX	EEM	+0.16 [-0.19,+0.52]	+0.73	51%	79%/68%	+29.7%	Mar 2025	REVIEW
ODMAX	EEM	+0.11 [-0.29,+0.48]	+0.49	49%	91%/86%	+14.2%	Aug 2023	REVIEW
PDEZX	EEM	+0.19 [-0.44,+0.84]	+0.65	51%	102%/89%	+27.6%	Mar 2026	WATCH
PEIFX	EEM	+0.22 [-0.50,+0.90]	+0.73	54%	97%/85%	+22.9%	Jan 2020	REVIEW
PRMSX	EEM	-0.10 [-0.59,+0.40]	-0.46	50%	99%/99%	-9.6%	Feb 2011	REVIEW
REMIX	EEM	+0.11 [-0.66,+0.91]	+0.28	43%	42%/4%	+13.1%	Feb 2026	WATCH
SFENX	EEM	+0.08 [-0.39,+0.55]	+0.34	49%	95%/91%	+10.0%	Dec 2012	WATCH
TEDMX	EEM	+0.02 [-0.32,+0.40]	+0.10	54%	96%/94%	+2.7%	Aug 2015	WATCH
TEAFX	EEM	-0.18 [-0.63,+0.24]	-0.84	48%	73%/73%	-33.1%	Jul 2010	WATCH
VEIEX	EEM	-0.08 [-0.42,+0.32]	-0.38	52%	93%/92%	-6.8%	Dec 2012	REVIEW
VEMAX	EEM	-0.02 [-0.40,+0.37]	-0.09	54%	94%/92%	-1.5%	Apr 2017	REVIEW
WAEMX	EEM	+0.16 [-0.31,+0.63]	+0.70	56%	88%/77%	+40.9%	-	CLEAN
WIEMX	EEM	-0.09 [-0.68,+0.55]	-0.28	50%	87%/83%	-10.2%	-	CLEAN
■ EM HARD CCY DEBT								
OP00008T6N.EMB	EMB	-0.53 [-1.24,+0.11]	-1.23	44%	39%/19%	-14.3%	Feb 2026	WATCH
OP0001KL95EMB	EMB	-0.07 [-1.15,+0.99]	-0.12	41%	63%/23%	-1.3%	-	CLEAN
ABEMX	EMB	+0.07 [-0.44,+0.60]	+0.31	51%	151%/154%	+20.5%	Dec 2015	WATCH

Summary Results

Risk-adjusted statistics — grouped by asset class, 24m calibration excluded — page 4 of 5

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	First Alarm	Signal
DBLEX	EMB	-0.03 [-0.48,+0.39]	-0.13	48%	72%/61%	-2.3%	Feb 2025	REVIEW
EBNDX	EMB	-0.16 [-0.74,+0.49]	-0.53	58%	87%/92%	-28.9%	-	CLEAN
IEMIX*	EMB	+1.58 [-0.79,+6.78]	+0.91	75%	-	+17.4%	-	CLEAN
MSEDX*	EMB	-1.41 [-2.34,-0.60]	-2.58	35%	49%/93%	-18.6%	-	CLEAN
PEBIX	EMB	+0.11 [-0.20,+0.57]	+0.47	51%	91%/83%	+9.1%	Dec 2015	WATCH
PREMX	EMB	-0.01 [-0.34,+0.45]	-0.03	53%	90%/87%	-0.5%	Dec 2014	REVIEW
TGBAX	EMB	-0.17 [-0.65,+0.21]	-0.71	46%	58%/56%	-25.7%	Sep 2015	REVIEW
VWOB	EMB	+0.08 [-0.37,+0.48]	+0.30	48%	94%/91%	+1.5%	Jul 2024	WATCH
■ EM LOCAL CCY DEBT								
PFMIX	EMLC	+0.26 [-0.18,+0.66]	+1.05	49%	32%/9%	+51.4%	Nov 2017	WATCH
PRLAX	EMLC	+0.04 [-0.39,+0.49]	+0.16	49%	174%/160%	+12.1%	Jan 2015	REVIEW
■ GLOBAL EQUITY								
OP00008T6D.9	ACWI	+0.64 [-0.20,+1.61]	+1.49	55%	126%/71%	+59.2%	-	CLEAN
OP00008T6G.9	ACWI	-0.53 [-1.48,+0.26]	-1.23	39%	61%/61%	-24.9%	May 2024	WATCH
OP000116K1.9	ACWI	-1.15 [-2.00,-0.39]	-2.26	37%	64%/69%	-24.0%	-	CLEAN
OP0001RMIN.9	ACWI	-1.46 [-2.86,-0.35]	-2.35	32%	45%/41%	-26.6%	-	CLEAN
AIVSX	URTH	+0.50 [+0.03,+0.97]	+1.88	55%	99%/86%	+28.5%	May 2017	WATCH
CWGIX	ACWI	-0.09 [-0.52,+0.31]	-0.40	46%	95%/94%	-5.0%	Feb 2017	WATCH

Summary Results

Risk-adjusted statistics — grouped by asset class, 24m calibration excluded — page 5 of 5

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	First Alarm	Signal
GPROX	ACWI	-0.35 [-0.98,+0.30]	-1.27	51%	96%/108%	-27.8%	May 2023	REVIEW
HAINX	URTH	-0.74 [-1.32,-0.26]	-2.79	44%	83%/103%	-50.9%	Sep 2018	REVIEW
JGVAX*	URTH	-1.11 [-3.20,-0.89]	-1.67	26%	11%/100%	-55.9%	-	CLEAN
MGGIX	ACWI	+0.38 [-0.29,+1.09]	+1.62	59%	114%/96%	+94.9%	Apr 2024	WATCH
MGIAX	URTH	-0.18 [-0.65,+0.23]	-0.67	44%	85%/81%	-16.7%	Apr 2018	REVIEW
OPGIX	ACWI	-0.09 [-0.57,+0.42]	-0.36	48%	112%/118%	-16.6%	Apr 2013	WATCH
SGIGX	ACWI	-0.42 [-1.18,+0.33]	-1.21	44%	89%/94%	-15.9%	Feb 2024	REVIEW
TEDIX	ACWI	-0.16 [-0.74,+0.39]	-0.68	46%	75%/72%	-22.3%	Sep 2012	WATCH
VHGEX	ACWI	-0.14 [-0.61,+0.35]	-0.59	51%	102%/104%	-7.8%	Feb 2023	REVIEW

Multiple-Testing Caution | Key Observations

MULTIPLE TESTING: With 68 funds at h=19.81 (ARL=60m/fund), expected FALSE alarms = 13.6/year across the universe. GFC: 1 alarm(s) | COVID: 0 alarm(s) — treat as regime signals, not skill-loss signals.
* = Short monitoring history (<36m live CUSUM; treat with caution). Best IR: IEMIX (+1.58, t=+0.9). Weakest: 0P0001RMIN.SI (-1.46).
Clean (no alarm): VWELX, PTTRX, PRWCX, DODWX, LSBRX, JEMWX ...

Post-Alarm Analysis

Cumulative excess return in months following the FIRST alarm — page 1 of 4

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#AI	Sig%(all)
FCNTX	May 2011	+2.1%	+1.4%	+0.7%	+0.1%	-1.4%	FALSE	1	0%
DODGX	Dec 2011	+22.0%	+17.5%	+4.5%	+16.3%	+13.8%	FALSE	1	0%
AGTHX	Jan 2011	+0.2%	+4.2%	-4.0%	-3.6%	+0.5%	CORRECT	1	100%
OTCFX	Nov 2013	+6.6%	+4.1%	+2.4%	+2.5%	+2.3%	FALSE	1	0%
OAKIX	Jul 2014	+1.6%	-0.1%	+1.7%	-3.2%	+11.8%	FALSE	1	0%
TEDMX	Aug 2015	+16.9%	+10.5%	+6.4%	+13.4%	+14.1%	FALSE	1	0%
FMAGX	Dec 2011	+18.0%	+16.0%	+2.0%	+6.3%	+8.1%	FALSE	1	0%
SEQUX	Sep 2018	+6.0%	+4.1%	+1.9%	+1.1%	+7.1%	FALSE	1	0%
FLPSX	Oct 2010	+8.7%	+6.7%	+2.0%	+1.5%	-0.2%	FALSE	1	0%
NWFFX	Mar 2025	+24.2%	+33.1%	-8.9%	n/a	n/a	CORRECT	1	100%
PRMSX	Feb 2011	-1.3%	-1.3%	-0.0%	+2.6%	+2.1%	CORRECT	1	100%
ODMAX	Aug 2023	+6.6%	+13.6%	-7.0%	-18.5%	n/a	CORRECT	1	100%
PEIFX	Jan 2020	+10.7%	+28.6%	-18.0%	+8.6%	+15.3%	CORRECT	1	100%
LZOEX	Aug 2017	-9.6%	-1.6%	-8.0%	-5.7%	-23.8%	CORRECT	1	100%

Post-Alarm Analysis

Cumulative excess return in months following the FIRST alarm — page 2 of 4

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#AI	Sig%(all)
HLMEX	Aug 2018	-2.3%	-4.8%	+2.5%	-3.9%	+0.5%	FALSE	1	0%
MEMCX	Oct 2013	-3.7%	+1.0%	-4.6%	-3.0%	-6.3%	CORRECT	1	100%
VEIEX	Dec 2012	-5.2%	-3.6%	-1.6%	+2.6%	+2.8%	CORRECT	1	100%
VEMAX	Apr 2017	+16.8%	+19.5%	-2.7%	-0.9%	+0.9%	CORRECT	1	100%
NEWFX	Mar 2025	+24.2%	+33.1%	-8.9%	n/a	n/a	CORRECT	1	100%
TEMTX	Jul 2010	+20.3%	+15.9%	+4.4%	+2.1%	+31.9%	FALSE	1	0%
SFENX	Dec 2012	-2.9%	-3.6%	+0.7%	-6.4%	-8.3%	FALSE	1	0%
REMITX	Feb 2026	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
ABEMX	May 2024	+7.9%	+11.5%	-3.6%	+0.0%	n/a	CORRECT	1	100%
GERIX	Apr 2019	-14.7%	-14.2%	-0.5%	+7.4%	+9.8%	CORRECT	1	100%
HLEMTX	Oct 2025	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
BGEFTX	Oct 2025	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
MIEMTX	Sep 2012	+0.5%	+0.6%	-0.0%	+2.6%	+9.3%	CORRECT	1	100%
EEMAX	Dec 2018	+30.8%	+18.2%	+12.6%	+36.3%	+28.2%	FALSE	1	0%

Post-Alarm Analysis

Cumulative excess return in months following the FIRST alarm — page 3 of 4

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#AI	Sig%(all)
FSEAX	Oct 2016	+33.3%	+26.6%	+6.7%	+2.8%	+20.8%	FALSE	1	0%
MADCX	Dec 2008	+75.7%	+69.0%	+6.8%	+8.8%	+9.9%	FALSE	1	0%
PDEZX	Mar 2026	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
PEBIX	Dec 2015	+14.9%	+9.3%	+5.6%	+6.6%	+7.9%	FALSE	1	0%
PREMX	Dec 2014	+0.7%	+1.0%	-0.4%	+5.0%	+4.1%	CORRECT	1	100%
ABEMX	Dec 2015	+12.0%	+9.3%	+2.7%	+25.3%	+10.5%	FALSE	1	0%
TGBAX	Sep 2015	+0.7%	+15.9%	-15.2%	-6.6%	-5.0%	CORRECT	1	100%
DBLEX	Feb 2025	+6.6%	+11.8%	-5.2%	n/a	n/a	CORRECT	1	100%
VWOB	Jul 2024	+8.7%	+8.5%	+0.1%	n/a	n/a	FALSE	1	0%
PFMIX	Nov 2017	+1.7%	-6.9%	+8.6%	+11.5%	+13.6%	FALSE	1	0%
PRLAX	Jan 2015	-25.3%	-15.2%	-10.1%	+13.9%	+34.5%	CORRECT	1	100%
CWGIX	Feb 2017	+20.7%	+19.0%	+1.7%	-2.2%	-2.4%	FALSE	1	0%
TEDIX	Sep 2012	+19.4%	+18.1%	+1.3%	+1.6%	+2.3%	FALSE	1	0%
VHGEX	Feb 2023	+21.6%	+23.3%	-1.7%	-6.3%	-15.7%	CORRECT	1	100%

Post-Alarm Analysis

Cumulative excess return in months following the FIRST alarm — page 4 of 4

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
OPGIX	Apr 2013	+24.7%	+15.1%	+9.6%	+1.2%	+25.1%	FALSE	1	0%
GPROX	May 2023	+9.9%	+23.6%	-13.8%	-22.6%	-48.0%	CORRECT	1	100%
SGIGX	Feb 2024	+13.1%	+15.3%	-2.1%	-6.4%	n/a	CORRECT	1	100%
MGGIX	Apr 2024	+23.6%	+12.0%	+11.5%	-18.5%	n/a	FALSE	1	0%
AIVSX	May 2017	+12.4%	+11.5%	+0.9%	-0.4%	+2.8%	FALSE	1	0%
MGIAX	Apr 2018	+6.5%	+7.2%	-0.6%	+3.7%	-9.6%	CORRECT	1	100%
HAIX	Sep 2018	-3.7%	+2.3%	-6.0%	-14.1%	-18.3%	CORRECT	1	100%
OP00008T6G.S	May 2024	+28.3%	+13.2%	+15.1%	-0.2%	n/a	FALSE	1	0%
OP00008T6N.S	Feb 2026	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a

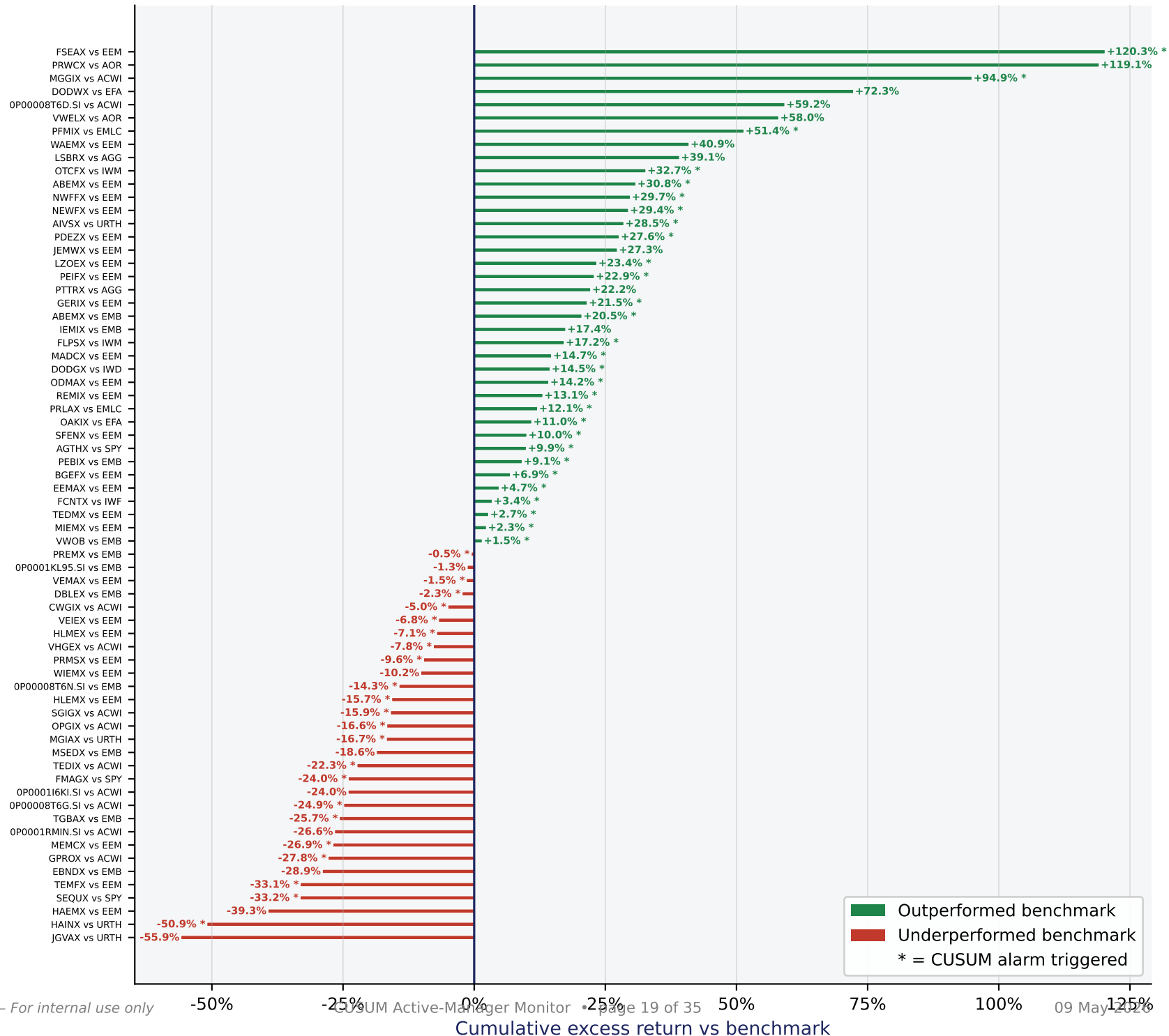
No alarm: VWELX, PTTRX, PRWCX, DODWX, LSBRX, JEMWX, HAEMX, WIEMX, WAEMX, EBNDX, IEMIX, MSIDX, JGVAX, OP0001I6KI.SI, OP00008T6D.SI, OP0001RMIN.SI, OP0001KL95.SI

How to Read This Table

+12M/+24M/+36M: cumulative excess return (fund minus benchmark) after the FIRST alarm.
1st Sig: CORRECT if fund underperformed after first alarm (signal validated). FALSE = false positive.
Sig%(all): % of ALL alarms where fund underperformed at +12M. >60% = reliable signal. 50% = coin flip.
At h=19.81, false-alarm rate = 1/60m per fund. Across a 67-fund universe, expect ~13 false alarms/year.

Cumulative Excess Returns — Ranked

Total return of each fund vs its benchmark (Jan 2005 – present)



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [1 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

FCNTX vs IWF

DODGX vs IWD



AGTHX vs SPY

OTCFX vs IWM



OAKIX vs EFA

TEDMX vs EEM

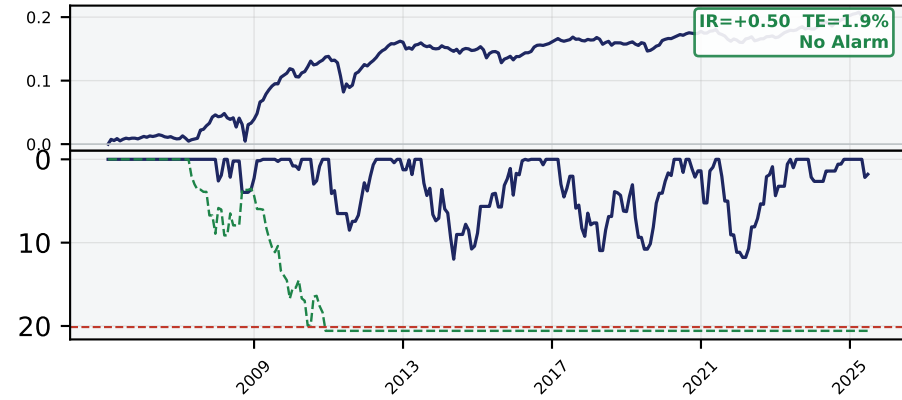


Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [2 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

VWELX vs AOR

PTTRX vs AGG



PRWCX vs AOR

FMAGX vs SPY



SEQUX vs SPY

FLPSX vs IWM



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [3 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

DODWX vs EFA

LSBRX vs AGG



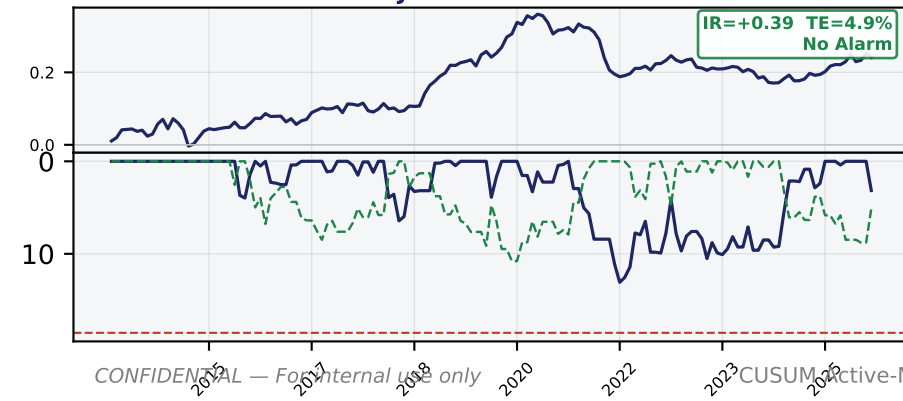
NWFFX vs EEM

PRMSX vs EEM



JEMWX vs EEM

ODMAX vs EEM



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [4 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

PEIFX vs EEM

LZOEX vs EEM



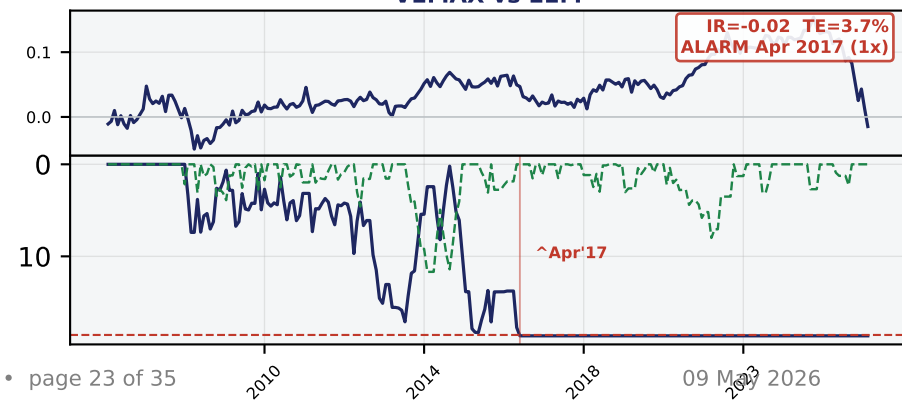
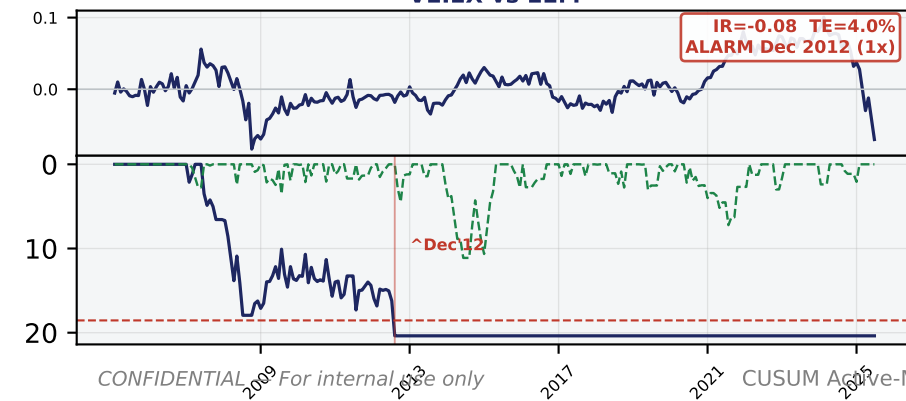
HLMEX vs EEM

MEMCX vs EEM



VEIEX vs EEM

VEMAX vs EEM



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [5 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

NEWFX vs EEM



HAEMX vs EEM



TEMFx vs EEM



SFENX vs EEM



REMIx vs EEM



ABEMX vs EEM



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [6 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

GERIX vs EEM



HLEMX vs EEM



BGEFX vs EEM



WIEMX vs EEM



WAEMX vs EEM



MIEMX vs EEM

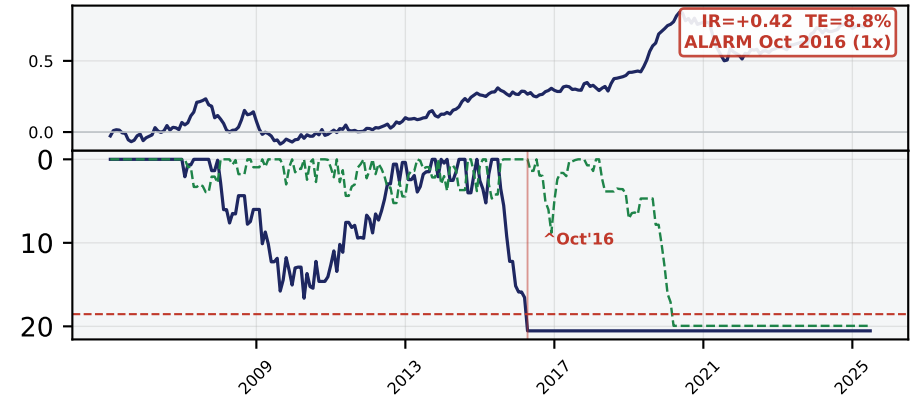


Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [7 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

EEMAX vs EEM

FSEAX vs EEM



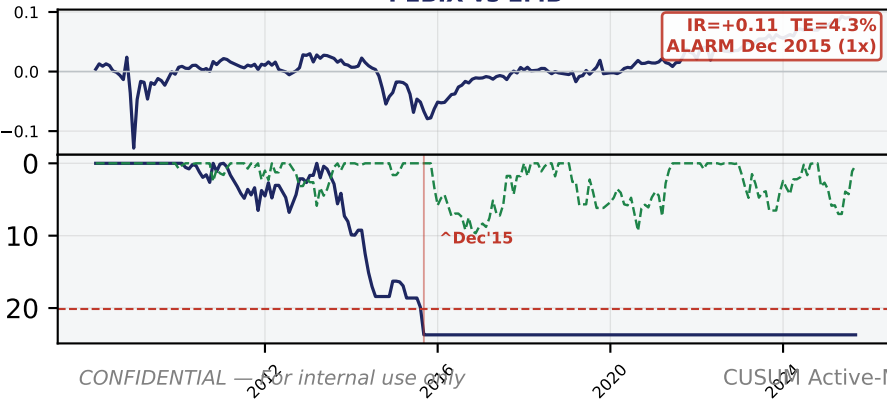
MADCX vs EEM

PDEZX vs EEM



PEBIX vs EMB

EBNDX vs EMB



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [8 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

PREMX vs EMB



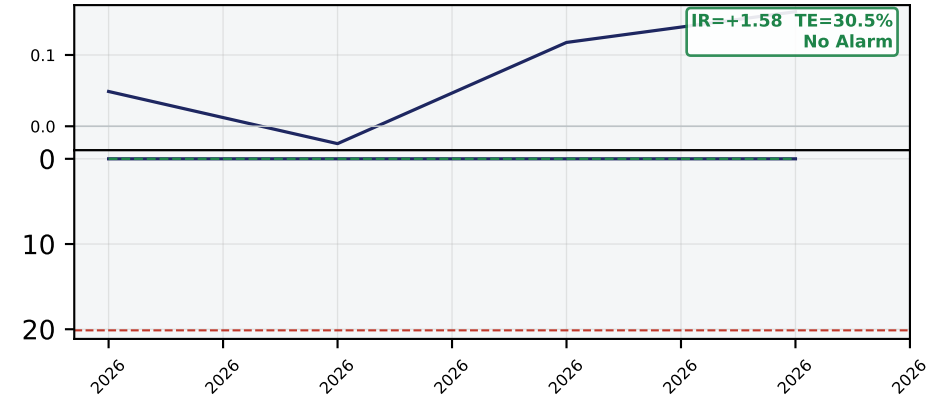
ABEMX vs EMB



TGBAX vs EMB



IEMIX vs EMB



DBLEX vs EMB



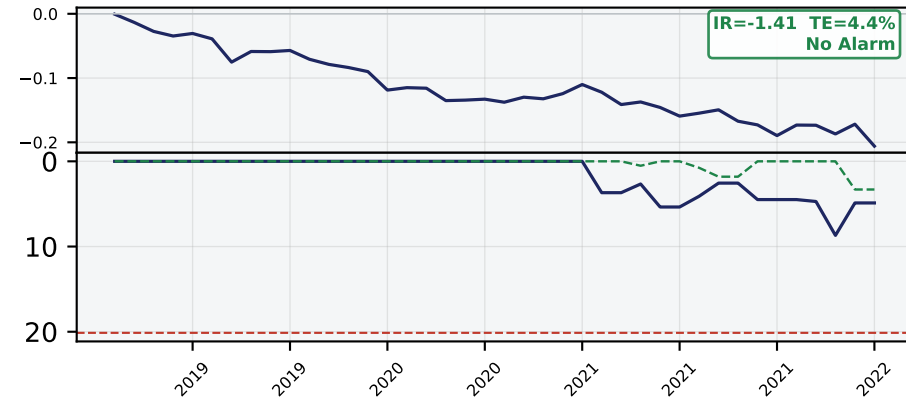
VWOB vs EMB



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [9 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

MSEDX vs EMB



PFMIX vs EMLC



PRLAX vs EMLC



CWGIX vs ACWI



TEDIX vs ACWI



VHGX vs ACWI



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [10 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

OPGIX vs ACWI



GPROX vs ACWI



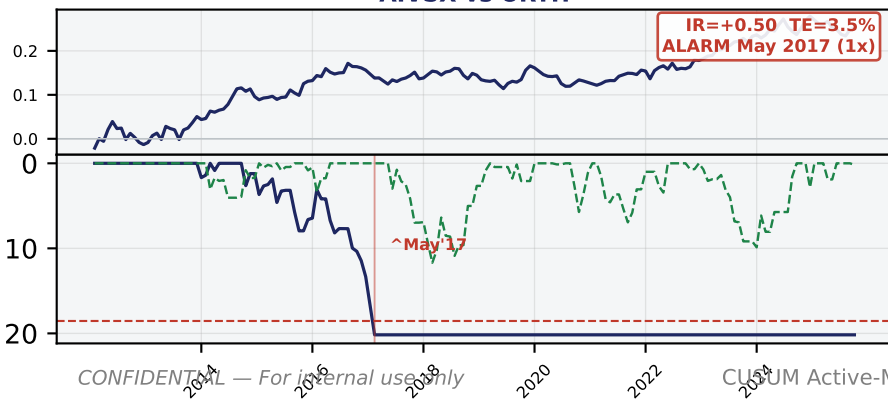
SGIGX vs ACWI



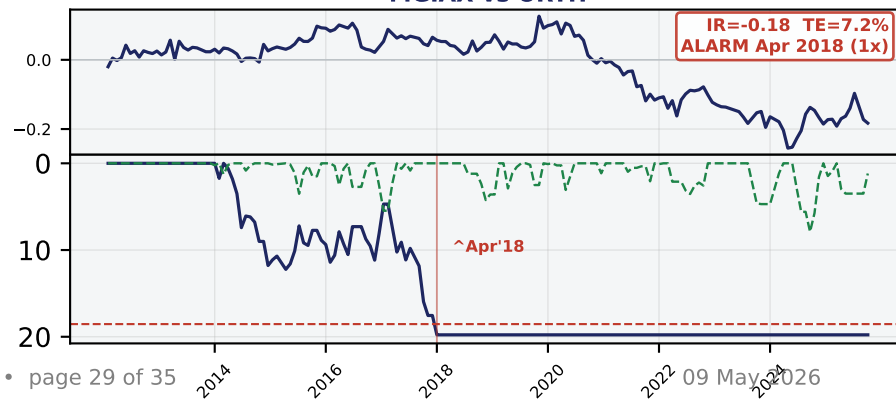
MGGIX vs ACWI



AIVSX vs URTH



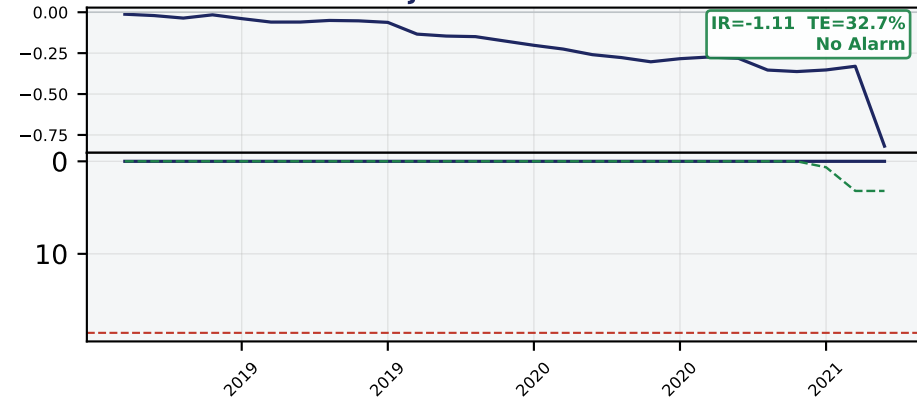
MGIAX vs URTH



Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [11 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

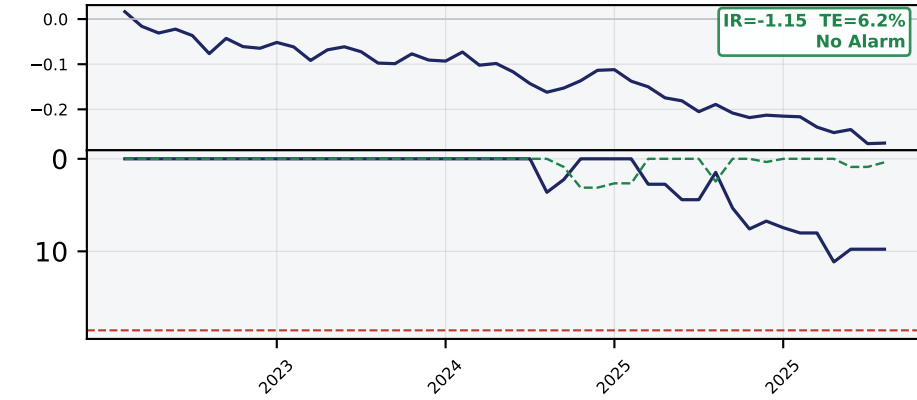
JGVAX vs URTH



HAINX vs URTH



OP0001I6KI.SI vs ACWI



OP00008T6G.SI vs ACWI



OP00008T6D.SI vs ACWI



OP0001RMIN.SI vs ACWI

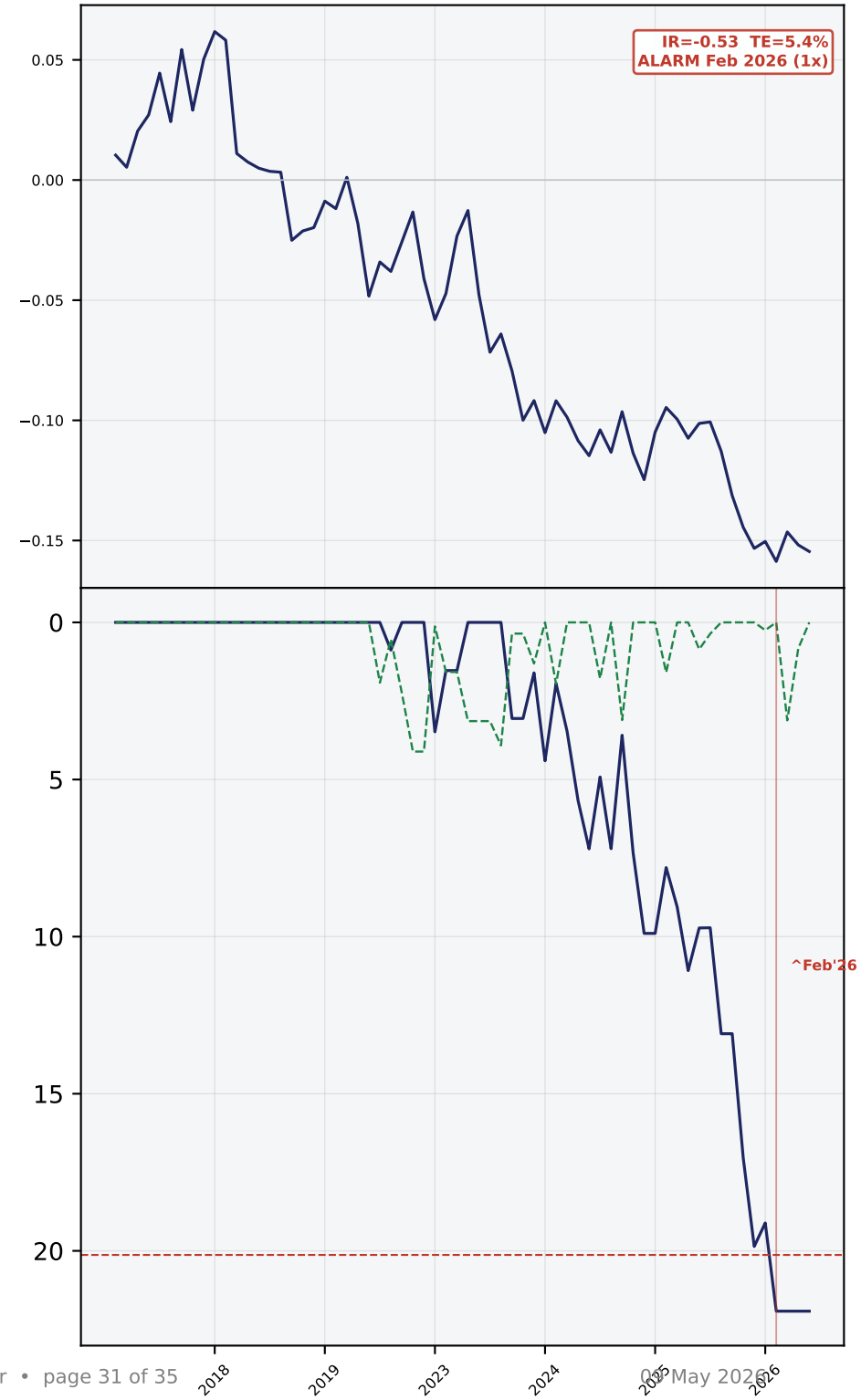
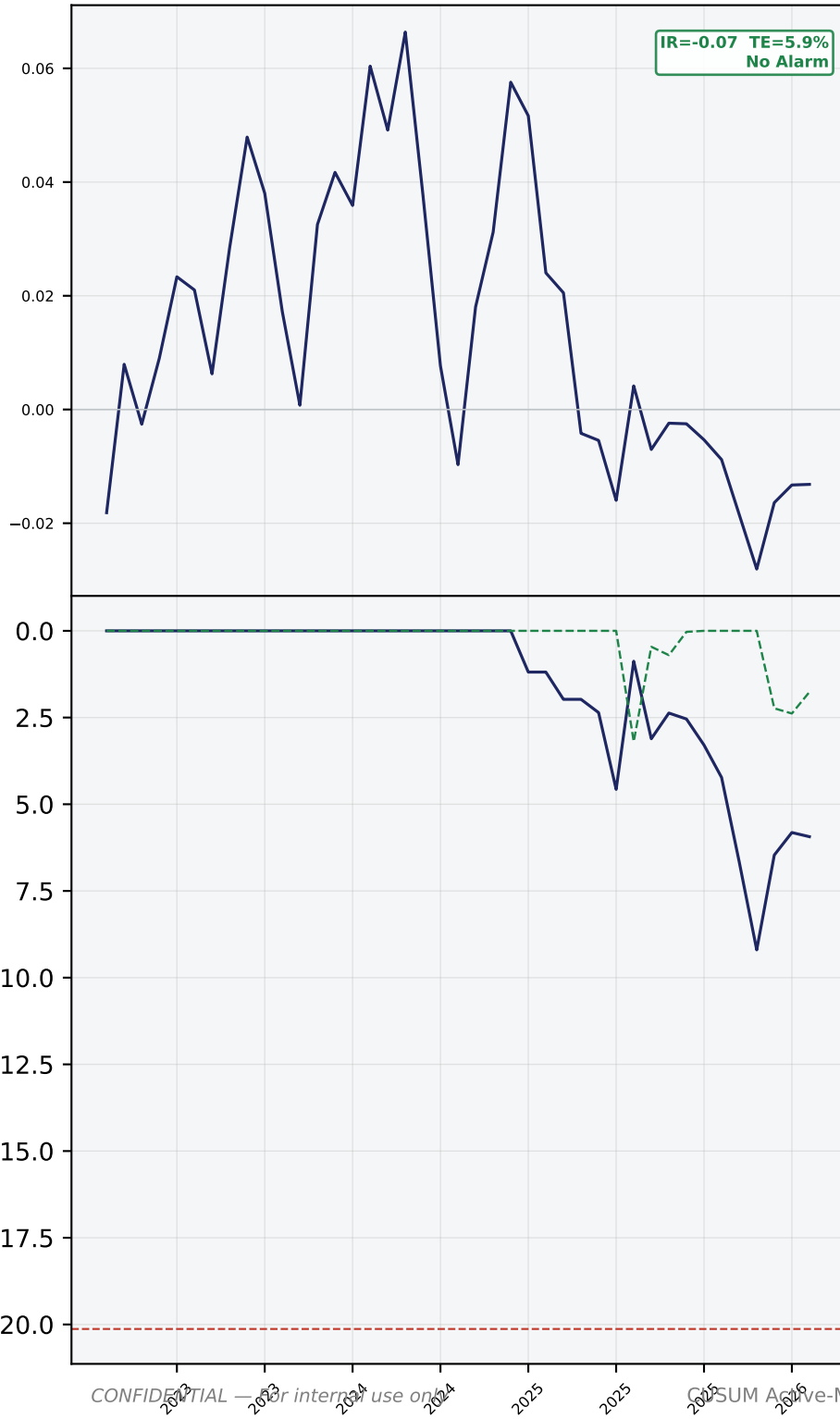


Appendix — Alarm frequency (informal IR proxy, not a calibrated metric) [12 / 12]

Multiple alarm markers visible only when rolling_restart=True. Default (paper-faithful) mode produces at most one alarm per fund.

0P0001KL95.SI vs EMB

0P00008T6N.SI vs EMB



Conclusions & Action Items

Alarm classification and recommended actions as of 09 May 2026 — page 1 of 4

Fund	Bench	Asset Class	IR	t-stat	#Alarms	First Alarm	Post +12M	Classification
AGTHX	SPY	US Large Blend	+0.10	+0.4	1	Jan 2011	-4.0%	REVIEW
NWFFX	EEM	Emerging Markets	+0.16	+0.7	1	Mar 2025	-8.9%	REVIEW
PRMSX	EEM	Emerging Markets	-0.10	-0.5	1	Feb 2011	-0.0%	REVIEW
ODMAX	EEM	Emerging Markets	+0.11	+0.5	1	Aug 2023	-7.0%	REVIEW
PEIFX	EEM	Emerging Markets	+0.22	+0.7	1	Jan 2020	-18.0%	REVIEW
LZOEX	EEM	Emerging Markets	+0.15	+0.7	1	Aug 2017	-8.0%	REVIEW
MEMCX	EEM	Emerging Markets	-0.32	-1.5	1	Oct 2013	-4.6%	REVIEW
VEIEX	EEM	Emerging Markets	-0.08	-0.4	1	Dec 2012	-1.6%	REVIEW
VEMAX	EEM	Emerging Markets	-0.02	-0.1	1	Apr 2017	-2.7%	REVIEW
NEWFX	EEM	Emerging Markets	+0.16	+0.7	1	Mar 2025	-8.9%	REVIEW
ABEMX	EEM	Emerging Markets	+0.24	+1.0	1	May 2024	-3.6%	REVIEW
GERIX	EEM	Emerging Markets	+0.23	+1.0	1	Apr 2019	-0.5%	REVIEW
MIEMX	EEM	Emerging Markets	+0.02	+0.1	1	Sep 2012	-0.0%	REVIEW
PREMX	EMB	EM Hard CCY Debt	-0.01	-0.0	1	Dec 2014	-0.4%	REVIEW

Conclusions & Action Items

Alarm classification and recommended actions as of 09 May 2026 — page 2 of 4

Fund	Bench	Asset Class	IR	t-stat	#Alarms	First Alarm	Post +12M	Classification
TGBAX	EMB	EM Hard CCY Debt	-0.17	-0.7	1	Sep 2015	-15.2%	REVIEW
DBLEX	EMB	EM Hard CCY Debt	-0.03	-0.1	1	Feb 2025	-5.2%	REVIEW
PRLAX	EMLC	EM Local CCY Debt	+0.04	+0.2	1	Jan 2015	-10.1%	REVIEW
VHGX	ACWI	Global Equity	-0.14	-0.6	1	Feb 2023	-1.7%	REVIEW
GPROX	ACWI	Global Equity	-0.35	-1.3	1	May 2023	-13.8%	REVIEW
SGIGX	ACWI	Global Equity	-0.42	-1.2	1	Feb 2024	-2.1%	REVIEW
MGIAX	URTH	Global Equity	-0.18	-0.7	1	Apr 2018	-0.6%	REVIEW
HAIX	URTH	Global Equity	-0.74	-2.8	1	Sep 2018	-6.0%	REVIEW
FMAGX	SPY	US Large Blend	-0.22	-1.0	1	Dec 2011	+2.0%	WATCH
SEQUX	SPY	US Large Blend	-0.22	-1.0	1	Sep 2018	+1.9%	WATCH
FCNTX	IWF	US Large Growth	+0.04	+0.2	1	May 2011	+0.7%	WATCH
DODGX	IWD	US Large Value	+0.14	+0.7	1	Dec 2011	+4.5%	WATCH
OTCFX	IWM	US Small Cap	+0.32	+1.5	1	Nov 2013	+2.4%	WATCH
FLPSX	IWM	US Small/Mid Cap	+0.09	+0.4	1	Oct 2010	+2.0%	WATCH

Conclusions & Action Items

Alarm classification and recommended actions as of 09 May 2026 — page 3 of 4

Fund	Bench	Asset Class	IR	t-stat	#Alarms	First Alarm	Post +12M	Classification
OAKIX	EFA	Intl Developed	+0.06	+0.3	1	Jul 2014	+1.7%	WATCH
TEDMX	EEM	Emerging Markets	+0.02	+0.1	1	Aug 2015	+6.4%	WATCH
HMEX	EEM	Emerging Markets	-0.06	-0.3	1	Aug 2018	+2.5%	WATCH
TEAFX	EEM	Emerging Markets	-0.18	-0.8	1	Jul 2010	+4.4%	WATCH
SFENX	EEM	Emerging Markets	+0.08	+0.3	1	Dec 2012	+0.7%	WATCH
REMX	EEM	Emerging Markets	+0.11	+0.3	1	Feb 2026	n/a	WATCH
HLEMX	EEM	Emerging Markets	-0.42	-1.1	1	Oct 2025	n/a	WATCH
BGEFX	EEM	Emerging Markets	+0.05	+0.2	1	Oct 2025	n/a	WATCH
EEMAX	EEM	Emerging Markets	+0.04	+0.2	1	Dec 2018	+12.6%	WATCH
FSEAX	EEM	Emerging Markets	+0.42	+1.9	1	Oct 2016	+6.7%	WATCH
PDEZX	EEM	Emerging Markets	+0.19	+0.6	1	Mar 2026	n/a	WATCH
PEBIX	EMB	EM Hard CCY Debt	+0.11	+0.5	1	Dec 2015	+5.6%	WATCH
ABEMX	EMB	EM Hard CCY Debt	+0.07	+0.3	1	Dec 2015	+2.7%	WATCH
VWOB	EMB	EM Hard CCY Debt	+0.08	+0.3	1	Jul 2024	+0.1%	WATCH

Conclusions & Action Items

Alarm classification and recommended actions as of 09 May 2026 — page 4 of 4

Fund	Bench	Asset Class	IR	t-stat	#Alarms	First Alarm	Post +12M	Classification
OP00008T6N.SMB		EM Hard CCY Debt	-0.53	-1.2	1	Feb 2026	n/a	WATCH
PFMIX	EMLC	EM Local CCY Debt	+0.26	+1.1	1	Nov 2017	+8.6%	WATCH
CWGIX	ACWI	Global Equity	-0.09	-0.4	1	Feb 2017	+1.7%	WATCH
TEDIX	ACWI	Global Equity	-0.16	-0.7	1	Sep 2012	+1.3%	WATCH
OPGIX	ACWI	Global Equity	-0.09	-0.4	1	Apr 2013	+9.6%	WATCH
MGGIX	ACWI	Global Equity	+0.38	+1.6	1	Apr 2024	+11.5%	WATCH
AIVSX	URTH	Global Equity	+0.50	+1.9	1	May 2017	+0.9%	WATCH
OP00008T6G.ACWI		Global Equity	-0.53	-1.2	1	May 2024	+15.1%	WATCH
MADCX	EEM	Emerging Markets	+0.12	+0.6	1	Dec 2008	+6.8%	REGIME

No alarm (17 funds): VWELX, PTTRX, PRWCX, DODWX, LSBRX, JEMWX, HAEMX, WIEMX, WAEMX, EBNDX, IEMIX, MSEDX, JGVAX, OP0001I6KI.SI, OP00008T6D.SI, OP0001RMIN.SI, OP0001KL95.SI

Statistical Caveats

- Multiple testing: 68 funds × ARL=60m → ~13.6 expected false alarms/year.
- GFC: 1 alarm(s) | COVID: 0 alarm(s)— regime signals, not skill loss.
- Funds marked (*) have <36m of live CUSUM— treat with caution.
- An alarm triggers a structured REVIEW, NOT automatic termination.

Recommended Next Steps

- Schedule IC review for all 22 REVIEW-classified manager(s).
- Place WATCH-classified managers on heightened monitoring.
- Confirm REGIME alarms are attributable to market stress.
- Verify short-history (*) fund data against official NAV sources.
- Re-run this report monthly after each month-end NAV update.